

3.5 Fundamentals of computer networks

Explain the following physical network topologies: star, bus.

Network topologies

Introduction

You have learned about LANs and WANs. You have learned that networks can be connected by wires, or by wireless connections. In this section you will learn about the different shapes that networks can have. We call these network topologies.

Network topology

Networks are made by connecting computers together. The connections are made using a transmission medium. That can be wired or wireless. The network of connections can make different shapes. The different shapes are called network topologies.

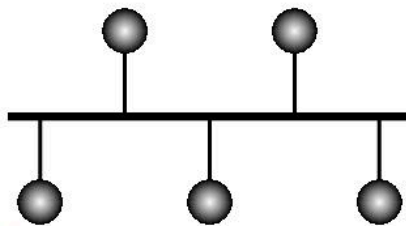
Bus topology

A bus topology has a single main cable that transmits all the signals. This main cable is called the bus. Every computer is connected to the bus by a short connecting wire. The signals pass up and down the wire. At each end of the wire is a terminator. The terminator turns the signal around and sends it back up the wire.

The big advantage of the bus topology is that it is simple and inexpensive to set up. There is only one cable. Any computer on the network can connect through that single cable. If any computer breaks down the network will keep running.

One disadvantage of the bus topology is that the cable can be overloaded with signals. If the network gets large or busy the signal becomes slow and unreliable. Signals can interfere with each other. This is called a collision.

Other disadvantages also arise from the reliance on a single cable. The whole network will fail if that cable fails. And there may be security concerns, as every user shares the same transmission medium.



↑ Simple bus network

Star topology

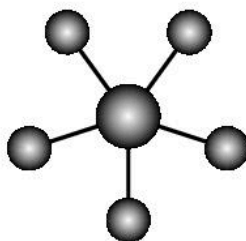
A star topology provides a different connection for every computer on the network. Here is a general diagram of the star topology.

One advantage in this system is that each computer has its own cable, so the network does not become overloaded. A fault in one part of the network, or one stretch of cabling, is easier to find and fix. It will not usually affect the rest of the network.

The star topology has some disadvantages. It uses more cabling than the bus topology, which is an additional expense. It contains a central 'switch' (see below) which is another additional expense. It is therefore more expensive to set up than a bus topology. Although cable failure will not disable the network, failure of the central switch stops the network from operating.

In a star network all communication passes through the centre of the star.

The element at the centre of the star is called a switch; this is a piece of equipment that receives signals from each computer in the network. The

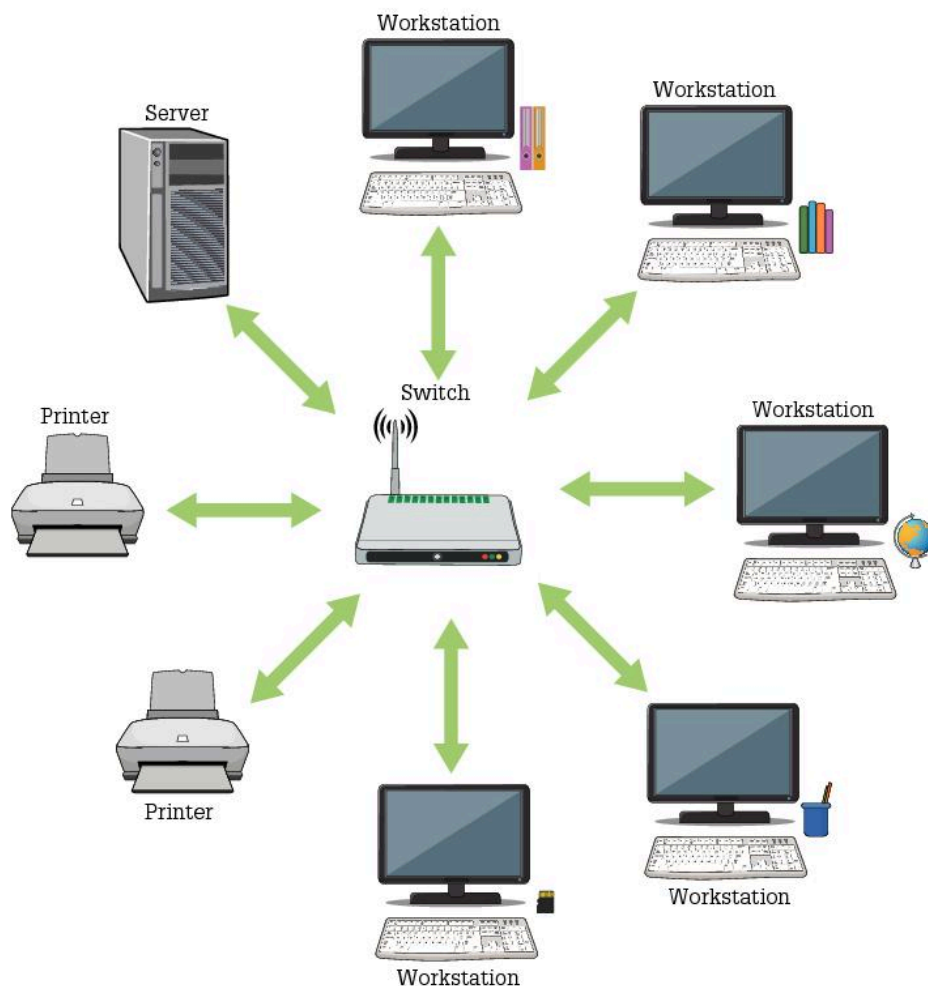


↑ Star topology

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switch contains a processor that examines each signal and only sends the signal to one computer – the one that needs to receive it.

In many cases a star topology will include a server – this is a large computer that provides services and storage space to the other computers.



↑ Wireless star topology

In recent years, the cost of cabling has decreased. Wireless systems have become cheaper and more reliable. These changes mean that star topology is a good choice for modern networks. It is the most common topology for LAN networks nowadays.

Learning activity

Draw the bus and network topologies using a graphics package.

Test yourself

1. Which network topology uses less cabling than other topologies?
2. Which network topology is best for a large network with a lot of signals? Explain your answer.
3. We usually draw the shape of a network topology by showing the wire connections. What is the topology of a wireless network?
4. What is the difference between a hub and a switch?

Extension activity

Another topology in use today is a mesh network. This is a particularly secure form of network topology. Research the shape of the mesh topology and find out when it is used.